PQA-Net: Deep No Reference Point Cloud Quality Assessment via Multi-View Projection

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***Abstract*— Recently, 3D point cloud is becoming popular due to its capability to represent the real world for advanced content modality in modern communication systems. In view of its wide applications, especially for immersive communication towards human perception, quality metrics for point clouds are essential. Existing point cloud quality evaluations rely on a full or certain portion of the original point cloud, which severely limits their applications. To overcome this problem, we propose a novel deep learning-based no reference point cloud quality assessment method, namely PQA-Net. Specifically, the PQA-Net consists of a multi-view-based joint feature extraction and fusion (MVFEF) module, a distortion type identification (DTI) module, and a quality vector prediction (QVP) module. The DTI and QVP modules share the feature generated from the MVFEF module. By using the distortion type labels, the DTI and the MVFEF modules are first pre-trained to initialize the network parameters, based on which the whole network is then jointly trained to finally evaluate the point cloud quality. Experimental results on the Waterloo Point Cloud dataset show that PQA-Net achieves better or equivalent performance comparing with the state-of-the-art quality assessment methods. The code of the proposed model will be made publicly available to facilitate reproducible research https://github.com/qdushl/PQA-Net.**

***Index Terms*— No-reference point cloud quality assessment, deep neural network, multi-task learning, multi-view.**

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I. Introduction

WITH the increasing capability of 3D data acquisition devices, point clouds are becoming more and more popular in a wide range of applications, from manufacture and construction to 3D telepresence [1]–[3]. A 3D point cloud comprises a set of points consisting of geometry and attribute information [4], [5]. The geometry denotes the 3D space coordinates of the points, and the attribute usually includes color, reflectance or normal vector of the points [6], [7]. Therefore, a point cloud can represent an object or a scene accurately and completely by millions of points directly [8].

Due to the content complexity of actual 3D scene and the limitation of specific sensors [9], the acquired raw point clouds always contain various types of noise, which are harmful for further applications. In addition, because of the huge data volume [9] of the acquired point clouds, it is difficult to achieve real-time lossless transmission under limited bandwidth [10], [11]. Therefore, lossy compression must be conducted before transmission, and thus the compression distortion is inevitable. As mentioned, point clouds are subject to distortions induced by acquisition, processing, and compression, any of which may lead to quality degradation [12]. Consequently, how to assess the quality of point clouds and thus evaluate the performance of the corresponding acquisition, processing, and compression systems becomes a critical issue [13].

Similar to image/video quality assessment, point cloud quality assessment can also be classified into three categories, i.e., full-reference (FR), reduced-reference (RR), and no-reference (NR), based on the way of exploring the original point cloud. Specifically, FR requires full access to the original point cloud, and RR relies on certain portion of the original point cloud. Recently, major progresses in FR and RR quality assessment have been made (see Section II). However, in many point cloud application scenarios, the original point cloud cannot be obtained. Especially for some terminal devices with limited storage and transmission capabilities, FR and RR quality assessment are unable to be conducted. Estimating the point cloud quality without the original one, i.e., NR quality assessment, is important for practical application since it can be carried out readily. To the best of our knowledge, there is no NR quality assessment for point clouds by now.

Motivated by the great success of deep neural networks (DNNs) on image/video processing and analysis, we propose an NR quality assessment method for point clouds

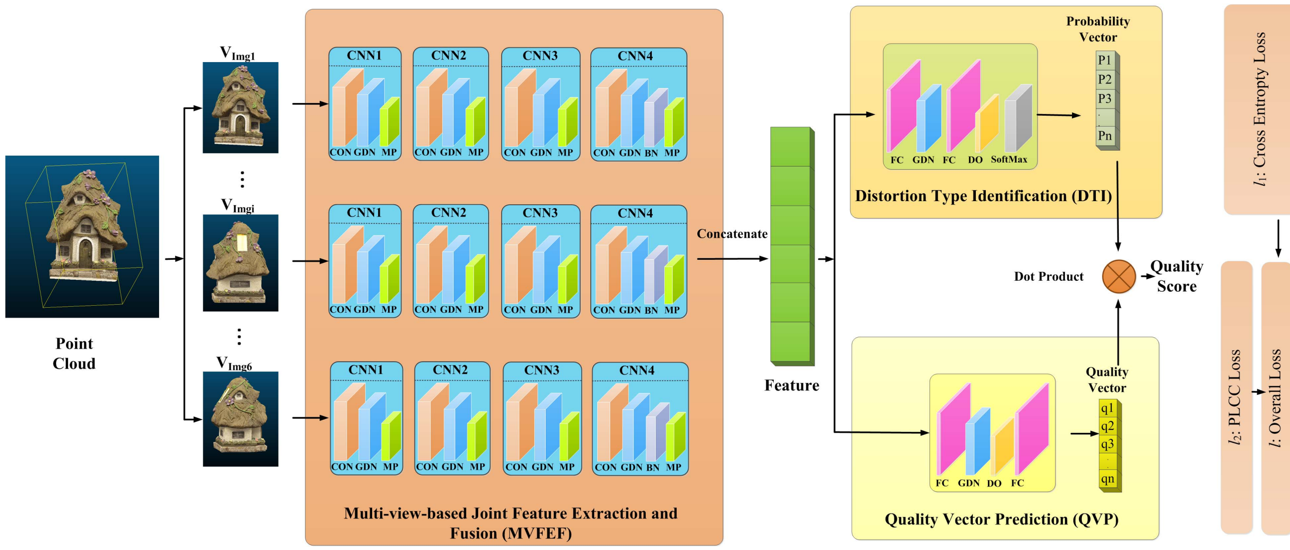


Fig. 1. Overview of the proposed PQA-Net, in which CNN denotes convolutional neural networks, CON denotes convolution layer and MP denotes maxpooling, GDN denotes the generalized divisive normalization, which is used to replace the activation function, BN is the batch normal unit, FC is fully connection module, DO is the dropout unit, l_1 is cross-entropy loss function, l_2 is defined based on person linear correlation coefficient (PLCC), and the overall loss function l is the linear weighting of l_1 and l_2.

based on DNNs. Usually, DNN needs a large dataset for training. Currently, the existing relatively large public point cloud quality assessment dataset, i.e., the Waterloo Point Cloud (WPC) dataset[1] [12], contains only 20 annotations and 720 degraded point clouds, which are not enough for directly training. To overcome this problem, inspired by [14] and [15] which deal with small sample dataset very well, we propose a DNN-based NR point cloud quality assessment method to estimate the quality of point clouds with a small sample data set. The NR point cloud quality assessment problem is divided into two sub-tasks. Sub-task I classifies a point cloud into a specific distortion type from a set of pre-defined categories. Sub-task II predicts the perceptual quality of the corresponding point cloud by taking the advantage of distortion information obtained from the distortion type classification task. The two sub-tasks are accomplished by two sub-networks with shared features, as shown in Fig. 1. The contributions of this paper are as follows.

- We propose to use DNN to assess the quality of point cloud. To the best of our knowledge, this is the first NR point cloud quality assessment method, which can be used directly without any original information.
- In the proposed network, the point cloud quality assessment task is divided into 2 cooperative sub-tasks, i.e., distortion type classification and quality prediction tasks, to deal with the small sample dataset efficiently.
- A multi-view projection strategy is adopted to effectively extract the feature of a point cloud comprehensively.
- Experimental results on the Waterloo dataset demonstrate that the proposed method achieves comparable or even better performance than existing state-of-the-art FR and NR methods.

The remainder of this paper is organized as follows. Brief review of related work for point cloud quality assessment is given in Section II. Then, the proposed method is presented in detail in Section III. Comprehensive experimental results and analyses are provided in Section IV. Finally, Section V concludes this paper.

[1]https://github.com/qdushl/Waterloo-Point-Cloud-Database

II. Related Work

Existing point cloud quality metrics can be broadly categorized as three-dimensional direct metrics and two-dimensional indirect metrics. Three-dimensional direct metrics rely on finding correspondences for all points or the relevant characteristics between the degraded point cloud and the original point cloud. While two-dimensional indirect metrics rely on the quality weight of the quality of multiple two-dimensional projections from the point cloud.

Usually, the Euclidean distance between corresponding points of the degraded and the original point clouds is used to measure the distortion, i.e., the point-to-point error adopted by MPEG [16]. However, it ignores the surface structures. Tian *et al.* [17] proposed point-to-plane distance to measure the geometric distortion, which is less dependent on a complex surface construction. Beside Euclidean distance, Alexiou and Ebrahimi [18] proposed a promising alternative objective quality metric for point clouds based on the angular similarity geometric distortion. Javaheri *et al.* [19] used the generalized Hausdorff distance to evaluate the geometry quality of point clouds. Meynet *et al.* [20] used the local curvature statistics to evaluate the quality of point clouds. Viola *et al.* [21] extracted color statistics such as histograms and correlograms to assess the quality of point cloud. Similarly, Meynet *et al.* [22] extracted a set of geometry and color features for each point based on a correspondence between the distorted and reference point clouds, then, proposed an FR quality metric as a linear combination of an optimal subset of the extracted features. Diniz *et al.* [23]–[25] proposed a local luminance pattern descriptor-based and local binary pattern descriptor-based distances to assess the perceived quality of the test point cloud. Besides, Yang *et al.* [26] constructed a local graph to aggregate color gradient moments and estimate the quality of point cloud. He also exploited multiscale potential energy discrepancy to measure point cloud geometry and color difference in [27]. Javaheri *et al.* [28] proposed a geometry quality metric which computes the *Mahalanobis* distance between a point on the reference point cloud and a distribution of points on a small region of the degraded one to measure the geometry quality.

Two-dimensional indirect metrics project a point cloud onto multiple 2D images from different viewpoint, and finally predict the quality by weighting the quality of these 2D images [29]. In this kind of methods, existing image distortion metrics such as peak signal-to-noise ratio (PSNR) [30], structural similarity (SSIM) [31], visual information fidelity in pixel domain (VIFP) [32] and multi-scale structural similarity (MS-SSIM) [33] are usually adopted directly to evaluate the quality of the projected 2D images. De Queiroz and Chou [34] applied the PSNR to guide rate distortion optimization in the codec design. In our previous work [35], we have derived a distortion metric by linearly combing the geometric distortion and the color distortion for point clouds, and also proposed a corresponding rate distortion optimization method for point cloud compression. Yang *et al.* [36] offered an objective FR metric via image feature aggregation by weighting global and local features extracted from two-dimensional color and depth images of all projection planes. Despite the state-of-the-art FR quality metrics, Alexiou and Ebrahimi [37] reduced the dependence on the original point cloud information to some extent, and proposed an RR quality metric which utilized a family of statistical dispersion measurements for the prediction of perceptual degradations. In our previous work [38], we also proposed an RR quality metric for point cloud based on the video-based point cloud coding (V-PCC) platform given by MPEG. All of the above methods either work on 3D or 2D rely heavily on the whole or part of original point cloud, that is to say, there is no NR quality assessment method for point clouds by now.

III. Proposed PQA-Net

First, we introduce the pre-processing method briefly. Then, we describe the details of the proposed PQA-Net. Finally, we wrap up this section by introducing the training and testing procedures for the proposed PQA-Net.

A. Pre-Processing

For a point cloud, one can only see its 2D appearances from a certain viewpoint at the same time. Thus, we first project a point cloud onto multiple 2D images, and use them to classify the distortion type and evaluate the quality. Specifically, to completely capture the characteristics of a point cloud, we project it onto six 2D image planes along the three orthogonal projections (up, down, left, right, front, and back). A point P_i in the point cloud is denoted by $P_i = (x_i, y_i, z_i, r_i, g_i, b_i)$ in which the first three elements (x_i, y_i, z_i) stand for the geometry coordinates and the last three elements (r_i, g_i, b_i) stand for red-green-blue

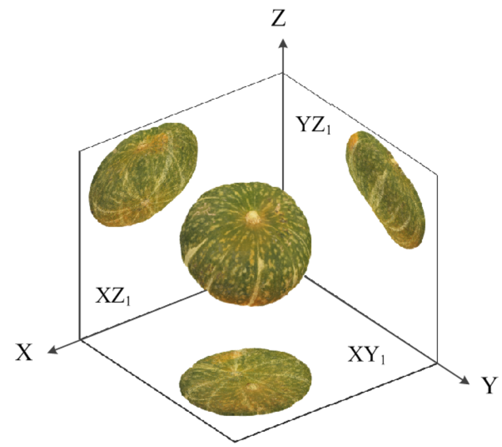


Fig. 2. Illustration of projecting a point cloud onto 2D image planes.

(RGB) color values. The six projected images are defined as $XY_1, XY_2, XZ_1, XZ_2, YZ_1, YZ_2$, where XY_1 and XY_2 are generated from XY-plane, XZ_1 and XZ_2 are generated from XZ-plane, and YZ_1 and YZ_2 are generated from YZ-plane. The pixel i of $XY_1, XY_2, XZ_1, XZ_2, YZ_1, YZ_2$ can be obtained as

$$\begin{cases} XY_1(i) = (x_i, y_i, r_{min(z_i)}, g_{min(z_i)}, b_{min(z_i)}), \\ XY_2(i) = (x_i, y_i, r_{max(z_i)}, g_{max(z_i)}, b_{max(z_i)}), \\ XZ_1(i) = (x_i, z_i, r_{min(y_i)}, g_{min(y_i)}, b_{min(y_i)}), \\ XZ_2(i) = (x_i, z_i, r_{max(y_i)}, g_{max(y_i)}, b_{max(y_i)}), \\ YZ_1(i) = (y_i, z_i, r_{min(x_i)}, g_{min(x_i)}, b_{min(x_i)}), \\ YZ_2(i) = (y_i, z_i, r_{max(x_i)}, g_{max(x_i)}, b_{max(x_i)}). \end{cases} \tag{1}$$

Take XY_1 plane as an example, x_i and y_i denote the coordinates of pixel i in the XY_1 plane. There maybe multiple points projected at the same position of XY_1 plane, therefore we select the point with the smallest Z-coordinate to be projected onto the XY_1 plane, that is $(r_{min(z_i)}, g_{min(z_i)}, b_{min(z_i)})$. The XY_2 plane is generated the same with XY_1 plane, except that it selects the color of the point with the maximal Z-coordinate in the projected XY_2 plane. The remaining projections are also generated similarly, as shown in Fig. 2. The resolution of all the projected images are set to be 1024×1024 as the side of the bounding box of the used point clouds is 1024.

B. Network Architecture

The projected 2D images are then fed into the proposed PQA-Net. We denote the input mini-batch training data by $\{(\mathbf{X}^{(k)}, \mathbf{p}^{(k)}, q^{(k)})\}_{k=1}^{K}$, where $\mathbf{X}^{(k)}$ is the k-th degraded point cloud represented by its corresponding six projected images, $\mathbf{p}^{(k)}$ is a multi-class indicator vector with only one entry activated to represent the ground truth distortion type, and $q^{(k)}$ is the mean of opinion score (MOS) [39][1] of the k-th input point cloud. The feature extractor is to transform the six projections $\mathbf{X}^{(k)}$ to a $64 \times 6 = 384$ dimension quality-related feature vector. It includes four lightweight convolutional neural

[1]MOS is a commonly used indicator of perceived media quality. It is defined as the average of opinion scores across subjects. The opinion score is a value in a predefined range (e.g. 0-100) that a subject assigns based on his opinion to the performance of a system.

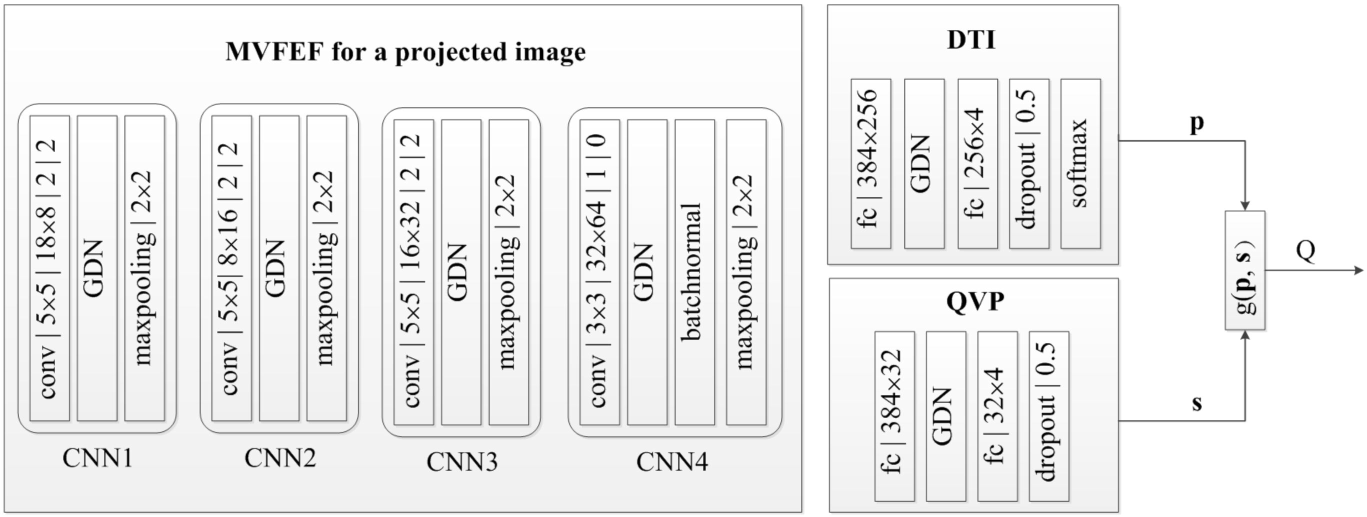


Fig. 3. Detailed architecture of the proposed PQA-Net, and denote the parameterizations of the convolutional layer as "height × width | input channel × output channel | stride | padding. DTI and QVP share features extracted from MVFEF and the outputs of the DTI and QVP are multiplied to calculate the quality of degraded point cloud."

networks (CNN) blocks, i.e., CNN1, CNN2, CNN3, and CNN4. The first three CNNs are all consisted of convolution (CON) layers, generalized divisive normalization (GDN)[2], and maxpooling (MP), whereas the CNN4 additionally adds a batch normal unit compared to the previous CNNs. The model parameters of the feature extractor are collectively denoted by $\mathbf{W}$, and the parameterizations of convolution, maxpooling, and connectivity from layer to layer are given in Fig. 3. Note that Fig. 3 only shows the detailed CNN structure for one projected image as an example. The parameters of the CNNs for the six projected images are shared. Since the point cloud has various shapes and sizes, to represent various point cloud shapes, the resolution of the projected images must be large enough, and the occupied effective pixels used to represent the point cloud in the projected images are also different. Therefore, we first detect the center pixel of the point cloud on the six projection planes, and then perform center cropping to obtain six cropped images with size of 235×235. As a result, we represent a point cloud by six 235×235 images with three channels (red, green, and blue), and then fed them into the feature extractor to generate a 384-dimensional feature vector.

The distortion identification and quality prediction subtasks are conducted by GDN, fully connection (FC), dropout (DO), and Softmax operations. Specifically, the architecture of DTI module is composed by two FC layers to compactly represent the input feature, one GDN transform to increase nonlinearity, and a dropout layer to reduce the probability of overfitting. We also adopt the softmax function to convert the unnormalized outputs of the last FC layer into $\mathfrak{D}$ distortion probability vectors $\hat{\mathbf{p}}^{(k)}(\mathbf{X}^{(k)}; \mathbf{W}, \mathbf{w}_1)$, where $\mathfrak{D}$ is the number of distortion types, and k is the index of the mini-batch. Since the cross entropy loss is very suitable for classification, and its convergence speed is fast [43], to classify the distortion type effectively and quickly, the cross entropy loss $l_1(\mathbf{X}^{(k)}; \mathbf{W}, \mathbf{w}_1)$ over the mini batch is used,

$$l_1(\mathbf{X}^{(k)}; \mathbf{W}, \mathbf{w}_1) = -\sum_{k=1}^{K}\sum_{i=1}^{D} p_i^{(k)} log[\hat{p}_i^{(k)}(\mathbf{X}^{(k)}; \mathbf{W}, \mathbf{w}_1)], \quad (2)$$

where $\mathbf{w}_1$ denotes the model parameters of the DTI module.

The QVP module for sub-task II has a similar structure to the distortion identification module, but lacks of the softmax unit, resulting in an architecture of two FCs, one GDN, and one DO. The QVP module takes in the features from MVFEF. Its output is then multiplied by the estimated distortion type probability vector $\hat{\mathbf{p}}^{(k)}$ to obtain the final quality score of a point cloud. The aim of this module is to predict the perceptual quality vector of $\mathbf{X}^{(k)}$ in the form of a scalar value $\hat{Q}^{(k)}$, whose parameters are collectively denoted by $\mathbf{w}_2$. The quality predictor produces a score vector $\hat{\mathbf{s}}^{(k)}$ whose i-th entry represents the perceptual quality score corresponding to the i-th distortion type. The final scalar value $\hat{Q}^{(k)}$ is computed by an inner-product of $\hat{\mathbf{p}}^{(k)}$ and $\hat{\mathbf{s}}^{(k)}$,

$$\hat{Q}^{(k)} = \hat{\mathbf{p}}^{(k)T}\hat{\mathbf{s}}^{(k)} = \sum_{i=1}^{\mathfrak{D}} \hat{p}_i^{(k)}\hat{s}_i^{(k)}. \quad (3)$$

Person linear correlation coefficient (PLCC) represents the linear correlation between the objective scores and the subjective scores. As it is a commonly-used evaluation criteria in the context of perceptual quality assessment, we defined loss function l_2 of Subtask II to improve the PLCC directly. Once the predicted score $\hat{Q}^{(k)}$ is obtained, the PLCC between the predicted scores and the ground truth $q^{(k)}$ in the mini-batch will be computed as

$$l_2(\mathbf{X}^{(k)}; \mathbf{W}, \mathbf{w}_1, \mathbf{w}_2) = \frac{\sum_{k=1}^{K}\left(\hat{Q}^{(k)} - \hat{Q}_m\right)\left(q^{(k)} - q_m\right)}{\sqrt{\sum_{k=1}^{K}\left(\hat{Q}^{(k)} - \hat{Q}_m\right)^2}\sqrt{\sum_{k=1}^{K}\left(q^{(k)} - q_m\right)^2}}, \quad (4)$$

[2]GDN is a differentiable transform that can be trained with any preceding or subsequent layers. It is inspired biologically [14]. Its effectiveness has been proven in image quality assessment [40], Gaussianizing image densities [41], and digital images compression [42]

(a) Downsample (b) Noise (c) G-PCC (d) V-PCC

Fig. 4. The distortion type of samples.

where $\hat{Q}_m$ and q_m denote the mean $\hat{Q}^{(k)}$ and $q^{(k)}$ across the mini-batch. The advantage of choosing the PLCC loss instead of the widely-used L1 or L2 loss functions is that human beings are more consistent with the rankings of perceptual quality rather than scores [44]. By taking the accuracy of distortion type classification task into account, we define the hybrid loss function of PQA-Net as

$$l(\mathbf{X}^{(k)}; \mathbf{W}, \mathbf{w}_1, \mathbf{w}_2) = l_1 - \lambda l_2, \tag{5}$$

where λ is a positive weight value to account for the scale difference between the DTI and QVP modules. The hybrid loss function contains the cross entropy (l_1) and PLCC (l_2), whose lower and higher values indicate better performance, respectively. Therefore, the PLCC (l_2) need to be subtracted from the cross-entropy (l_1) loss.

C. Training and Testing

The PQA models are trained on the newly and the existing relatively large collected data set [12] with two category labels, i.e., distortion types and MOS scores. PQA adopts a two-step strategy to train the multi-task neural network. The distortion type classification task and the quality prediction task share the extracted feature of MVFEF module, and the quality of a degraded point cloud is the dot product of the output of the DTI and QVP modules. At the first step, the model parameters $\mathbf{W}$ of the MVFEF module and the model parameters $\mathbf{w}_1$ of the DTI module are trained by minimizing the loss function

$$(\hat{\mathbf{W}}, \hat{\mathbf{w}}_1) = \arg\min l_1(\mathbf{X}^{(k)}; \mathbf{W}, \mathbf{w}_1). \tag{6}$$

In the second step, the model parameters $(\mathbf{W}, \mathbf{w}_1)$ of MVFEF and DTI modules are initialized as $(\hat{\mathbf{W}}, \hat{\mathbf{w}}_1)$, respectively, and the overall loss is minimized by jointly fine-tune the whole network. As a result, the optimal QVP module parameters $\widetilde{\mathbf{w}}_2$ and the optimal MVFEF and DTI module parameters $(\widetilde{\mathbf{W}}, \widetilde{\mathbf{w}}_1)$ can be obtained:

$$(\widetilde{\mathbf{W}}, \widetilde{\mathbf{w}}_1, \widetilde{\mathbf{w}}_2) = \arg\min l(\mathbf{X}^{(k)}; \mathbf{W}, \mathbf{w}_1, \mathbf{w}_2). \tag{7}$$

In the two-step training strategy, the first pre-training step allows us to train a quality-related feature extractor and distortion classifier, while the joint optimization step trains a quality predictor by taking the distortion types as a strong regularizer.

IV. Experimental Results and Analyses

In this section, we first describe the experimental setups including the implementation of PQA-Net and the Waterloo Point Cloud Sub-Dataset (WPCSD). We then did ablation study to confirm the influence of the λ and the DTI module. After that, we compared the proposed PQA-Net with optimal parameters with state-of-the-art FR and RR quality metrics and displayed the comparative visual results. Finally, we tested the performance of the proposed PQA-Net by selecting different loss functions and also tested its performance on other datasets.

A. Waterloo Point Cloud Sub-Dataset

We compared the proposed PQA-Net with classic and state-of-the-art FR and RR point cloud quality assessments metrics on the Waterloo Point Cloud Dataset [12] which covers various geometry and texture complexity. In the experiments, we selected 20 high-resolution point clouds with nearly pristine quality as the basis to construct the sub-dataset for PQA-Net. The 3D point clouds of Waterloo Point Cloud Dataset are constructed by using three steps. First of all, a single-lens-reflex camera and a turntable were employed to take multiview images of an object. Second, the Agisoft Photoscan [45] software was used to calibrate and align the captured multiview images and represent the object as a point cloud. Finally, each point cloud was normalized to a unit-cube with a step size of 0.001. One can refer to [12] for the detailed information of the point cloud generation procedure. We considered four distortion types, i.e., downsample distortion (60 degraded point clouds), Gaussian white noise (180 degraded point clouds), coding distortion induced by V-PCC (240 degraded point clouds with different geometry and color distortions) and G-PCC (180 degraded point clouds with different geometry and color distortions) respectively, as shown in Fig. 4. To augment the dataset, each point cloud was treated as 12 versions by rotating it to 12 different viewpoints, as shown in Fig. 5. The twelve viewpoints placed uniformly around the object based on the twelve polyhedron

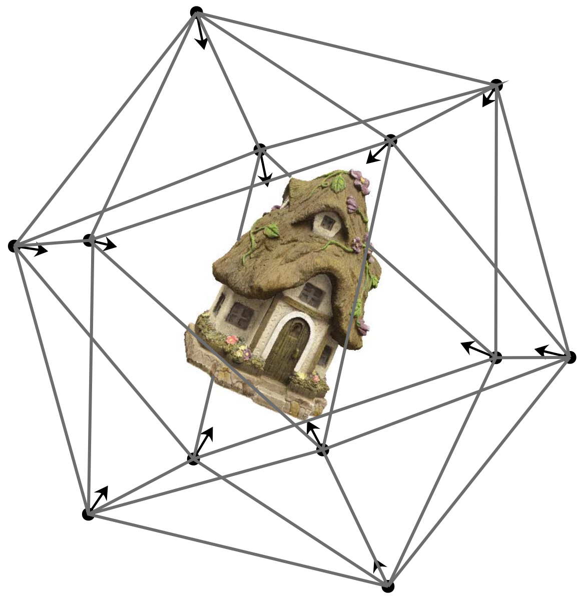


Fig. 5. Example of 12 viewpoints of one point cloud.

vertices of a regular icosahedron [46]. Then, we got (60 + 180 + 240 + 180) × 12 = 7920 distorted point clouds from twelve viewpoints, which is defined as Waterloo Point Cloud Sub-Dataset (WPCSD). Then the generated $7920 \times 6 = 47520$ projections were fed into the PQA-Net. The content as well as the number of points of each point cloud are shown in Fig. 6.

B. Experimental Setups

The Adam optimization with a mini batch of 30 was adopted for training. In both sub-tasks, we set the learning rate (α) to be 10^{-4} and subsequently decrease it by a factor of 0.1 until the loss converges. The remaining parameters of Adam were set by default [47]. The parameters β and γ in GDN [14] were clipped to nonnegative values after each update. Additionally, we enforced γ to be symmetric by averaging it with its transpose as suggested in [42]. In our work, the balance weight λ was set to be 10. To reduce computational complexity, as mentioned in part B of Section III, the six projected 1024×1024 images were cropped into a $235 \times 235 \times 3 \times 6$ matrix for each point cloud. Eighty percent of the point clouds in the dataset were randomly selected for training, while the remaining twenty percent were used for testing. The training set includes "*Bag*", "*Biscuits*", "*Cake*", "*Flowerpot*", "*Glasses_case*", "*Honeydew_melon*", "*House*", "*Litchi*", "*Pen_container*", "*Ping − pong_bat*", "*Puer_tea*", "*Pumpkin*", "*Ship*", "*Statue*", "*Stone*", "*Tool_box*", while the testing set includes "*Banana*", "*Cauliflower*", "*Mushroom*", "*Pineapple*".

C. Experimental Results

The common evaluation criterions, i.e., PLCC, SRCC, and KRCC, were adopted to compare the performance.

(1) Pearson linear correlation coefficient (PLCC) [48] is a non-parametric measure of the linear correlation, it can be computed as

$$PLCC = \frac{\sum_i (Q_i - Q_m)\left(\hat{Q}_i - \hat{Q}_m\right)}{\sqrt{\sum_i (Q_i - Q_m)^2}\sqrt{\sum_i \left(\hat{Q}_i - \hat{Q}_m\right)^2}}, \tag{8}$$

where Q_i and $\hat{Q}_i$ denotes the ground truth MOS and the predicted MOS of the i-th point cloud, respectively.

(2) Spearman's rank-order correlation coefficient (SRCC) [49] is another non-parametric measure for quality assessment field and defined as

$$SRCC = 1 - \frac{6\sum_i d_i^2}{I\left(I^2 - 1\right)}, \tag{9}$$

where I is the number of the tested point clouds and d_i is the rank difference between the ground truth MOS and the predicted MOS of the i-th point cloud. SRCC is independent of monotonic mappings.

(3) Kendall's rank-order correlation coefficient (KRCC) [50] is aimed to evaluate the association between two ordinal (two ranked variables, not necessarily intervals) variables and computed as

$$KRCC = \frac{N_c - N_d}{\frac{1}{2}N(N-1)}, \tag{10}$$

where N_c and N_d are the numbers of concordant (of consistent rank order) and discordant (of inconsistent rank order) pairs in the data set, respectively, and N is the number of elements of the variables.

Besides the above criterions, a perceptually weighted rank correlation (PWRC) indicator, which rewards the capability of correctly ranking high-quality images and suppresses the attention toward insensitive rank mistakes [51], is also used to compare the performance of different quality assessment algorithms.

In addition, we also computed Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) between the ground truth MOS and the predicted MOS of tested point cloud to further compare the accuracy. To compare with the ground truth MOS, the raw model predictions are mapped to the MOS scale by using a four-parameter logistic function.

1) Results Without Distortion Type Classification: To illustrate the importance of the distortion type classification sub-task, we tested the results without DTI module (namely PQA_W/O_DTI) and compared the result with our proposed PQA-Net. To make a fair comparison, all the training and testing point clouds are the same for PQA_W/O_DTI and PQA-Net. The network parameters are also the same except the parameters required by the DTI module. The results presented in Table I show that the DTI module can significantly improve the accuracy. The average PLCC and SRCC increment are 0.55 and 0.56, respectively, by adding the DTI module, indicating that the DTI module contributes a lot for handling the small sample dataset.

2) Experimental Results of Different lambda: The parameter λ in the loss function (5) also plays a key role in the proposed PQA-Net. It is used to balance the importance between the two sub-tasks. The performance of different λs are

(a) Bag, 1267845 points (b) Banana, 807184 points (c) Biscuits, 952479 points (d) Cake, 2486566 points

(e) Cauliflower, 1936627 points (f) Flowerpot, 2407154 points (g) Glasses_case, 716659 points (h) Honeydew_melon, 1431071 points

(i) House, 1568490 points (j) Litchi, 1039942 points (k) Mushroom, 1144603 points (l) Pen_container, 2878318 points

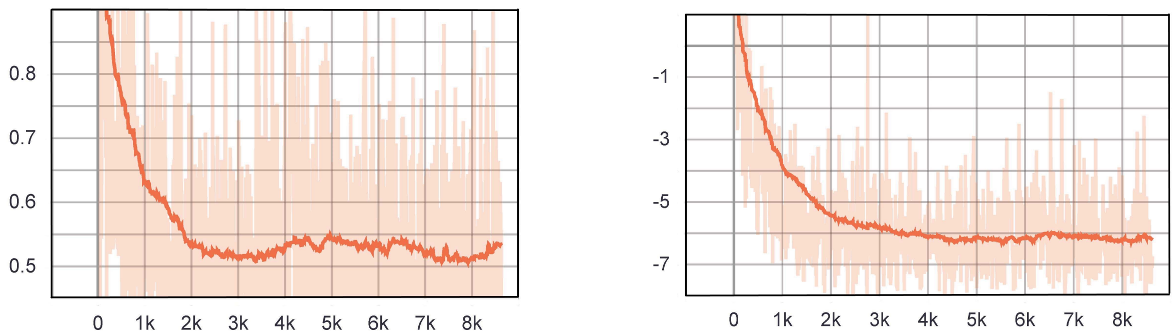
(m) Pineapple, 1628910 points (n) Ping-pong_bat, 703879 points (o) Puer_tea, 412009 points (p) Pumpkin, 1340343 points

(q) Ship, 684617 points (r) Statue, 1637577 points (s) Stone, 1086453 points (t) Tool_box, 1054211 points

Fig. 6. Snapshots and points number of the used point clouds.

shown in Table II, from which we can see that $\lambda = 10$ achieves the best. Besides, Fig. 7 shows the convergence curves of the training procedure with $\lambda = 10$.

3) Experimental Results on Testing Dataset of WPCSD: We compared the proposed PQA-Net with classic and state-of-the-art FR and RR point cloud quality metrics on WPCSD.



(a) Distortion type classifier loss

(b) Total loss

Fig. 7. The loss curves of $\lambda = 10$, where x-coordinate indicates step number and the y-coordinate indicates the loss value.

TABLE I
THE COMPARISON RESULT BETWEEN THE PQA_W/O_DTI AND PQA-NET

Method Name	PLCC	SRCC
PQA_W/O_DTI	0.15	0.13
PQA-Net	**0.70**	**0.69**
Accuracy Gain	0.55	0.56

TABLE II
INFLUENCE OF THE PARAMETER λ IN (5)

λ	PLCC	SRCC
1	0.55	0.50
5	0.58	0.56
10	**0.70**	**0.69**
15	0.70	0.67

The compared metrics are chosen to cover a diversity of design philosophies, including point-based, angular-based, projection-based. The most representative point-based point cloud quality metric is point-to-point PSNR only for geometry or color (Y component) [52], [53], [55], namely $PSNR_{MSE,p2po}$, $PSNR_{HF,p2po}$, and $PSNR_Y$, respectively. The geometry difference of one point in the reference point cloud and that of the nearest point (evaluated by Euclidean distance or Hausdorff distance) in the distorted point cloud is used to calculate the $PSNR_{MSE,p2po}$, $PSNR_{HF,p2po}$, while the color difference (MSE of Y component) of the matched points (evaluated by Euclidean distance) in the reference and the distorted point cloud is used to calculate the $PSNR_Y$. In addition, the point-to-plane PSNRs for geometry information [17], which are related with Euclidean and Hausdorff distances (namely $PSNR_{MSE,p2pl}$ and $PSNR_{HF,p2pl}$, respectively) were also compared. For the angular-based point cloud quality metrics, the angular similarity of associated points belonging to a reference and a point cloud under evaluation is used to assess the quality [18], e.g., AS_{Mean}, AS_{RMS}, and AS_{MSE}. For the projection-based point cloud quality metrics, the most essential idea is to estimate the average quality of the six projected images as the final point cloud quality, and the algorithms $SSIM$ [31], $MS\text{-}SSIM$ [33], and $VIFP$ [32] are usually employed to assess the projected image quality, namely $SSIM_p$, $MS\text{-}SSIM_p$, and $VIFP_p$, respectively. The state-of-the-art $GraphSIM$ was also compared. In this method a graph can be constructed in both reference and distorted point clouds to calculate the similarity index for quality assessment [26], all the parameters of $GraphSIM$ are following the published version [56]. We also compared the point cloud quality assessment method in [22] (namely, $PCQM$) and the method in [37] (namely, $PointSSIM$). The $PCQM$ uses an optimally weighted linear combination of geometry and color features to assess the point cloud quality. We followed the weight values recommended by [57] in our experiments. The $PointSSIM$ uses a wide set of features which represent local changes to estimate point cloud quality. We used the parameters recommended by [58] for $PointSSIM$. Beside the above FR quality metrics, a state-of-the-art PCM_{RR} [54] quality metric which evaluates the perceptual quality score by a linear combination of the differences of geometry, color and normal vector features were also compared. When implementing PCM_{RR}, all the parameters were set based on [59].

The SRCC, PLCC, KRCC, MAE, RMSE, and PWRC results are given in Table III. We should note that the quality metrics can either directly evaluate the quality or indirectly predict the quality. Therefore, the criterion (e.g. PLCC, SRCC, and KRCC) can be positive or negative. The closer the absolute value of PLCC, SRCC, or KRCC is to 1, the better the performance. These results provide some useful insights with respect to the approaches for point cloud quality assessment. First of all, the methods (i.e., $PSNR_{MSE,p2p0}$, $PSNR_{MSE,p2pl}$, $PSNR_{HF,p2p0}$, $PSNR_{HF,p2pl}$, AS_{Mean}, AS_{RMS}, and AS_{MSE}) that only consider the geometry structure do not perform well. The major reason is that these methods do not take the color information into consideration, which has been shown to be the major impact factor of point cloud quality [54]. Indeed, this is quite apparent from our test results, where even $PSNR_Y$, a crude quality metric performs significantly better than those only consider geometry information. $PCQM$ gives better performance than $PSNR_Y$ due to the additional geometry features. For $GraphSIM$, the PLCC and SRCC are 0.47 and 0.46 because it is very hard to construct the local graph accurately due to the complex geometric structure of point clouds. However, the $GraphSIM$ provides a new methodology for point cloud quality assessment, and maybe more suitable for other datasets. The PLCC and SRCC of the PCM_{RR} are 0.29 and 0.26 based on the optimal feature weights provided by [59]. The reason is that

TABLE III
ACCURACY COMPARISON WITH EXISTING FR AND RR POINT CLOUD QUALITY METRICS FOR DIFFERENT CONTENTS

Point Cloud Name	Method Type	Method	$\|PLCC\|$	$\|SRCC\|$	$\|KRCC\|$	MAE	RMSE	PWRC
Banana	FR	$PSNR_{MSE,p2po}$ [52]	0.73	0.64	0.47	11.64	14.73	0.68
		$PSNR_{MSE,p2pl}$ [17]	0.57	0.54	0.40	14.08	17.58	0.57
		$PSNR_{HF,p2po}$ [52]	0.35	0.07	0.02	16.83	20.13	0.21
		$PSNR_{HF,p2pl}$ [17]	0.35	0.08	0.04	17.00	20.38	0.09
		$PSNR_Y$ [53]	0.72	0.62	0.47	12.04	14.82	0.50
		AS_{Mean} [18]	0.40	0.39	0.30	15.58	19.64	0.31
		AS_{RMS} [18]	0.39	0.34	0.26	15.71	19.76	0.26
		AS_{MSE} [18]	0.39	0.34	0.26	15.70	19.74	0.26
		$SSIM_p$ [31]	0.72	0.77	0.60	11.51	14.88	0.56
		$MS\text{-}SSIM_p$ [33]	0.80	**0.82**	**0.65**	9.91	12.92	0.62
		$VIFP_p$ [32]	**0.81**	**0.82**	0.64	**9.70**	**12.50**	**0.70**
		$GraphSIM$ [26]	0.53	0.46	0.37	13.97	18.15	0.35
		$PCQM$ [22]	0.64	0.74	0.56	17.83	21.46	0.60
		$PointSSIM$ [37]	0.10	0.18	0.12	17.83	21.46	0.33
	RR	PCM_{RR} [54]	0.42	0.47	0.38	17.83	21.46	0.38
	NR	PQA-Net	0.53	0.52	0.39	13.96	18.17	0.38
Cauliflower	FR	$PSNR_{MSE,p2po}$ [52]	0.55	0.49	0.35	15.08	17.86	0.45
		$PSNR_{MSE,p2pl}$ [17]	0.40	0.30	0.23	15.68	19.50	0.30
		$PSNR_{HF,p2po}$ [52]	0.09	0.16	0.11	18.47	21.23	0.13
		$PSNR_{HF,p2pl}$ [17]	0.33	0.25	0.18	16.32	20.10	0.21
		$PSNR_Y$ [53]	0.56	0.54	0.40	13.99	17.65	0.43
		AS_{Mean} [18]	0.37	0.24	0.18	16.21	19.76	0.23
		AS_{RMS} [18]	0.37	0.24	0.18	16.21	19.76	0.30
		AS_{MSE} [18]	0.37	0.24	0.18	16.21	19.76	0.30
		$SSIM_p$ [31]	**0.80**	**0.77**	**0.61**	9.85	12.87	0.56
		$MS\text{-}SSIM_p$ [33]	0.76	0.74	0.58	10.92	13.74	**0.58**
		$VIFP_p$ [32]	**0.80**	0.75	0.59	**9.80**	**12.67**	**0.58**
		$GraphSIM$ [26]	0.61	0.59	0.43	13.15	16.81	0.44
		$PCQM$ [22]	0.67	0.69	0.52	18.22	21.32	0.56
		$PointSSIM$ [37]	0.36	0.21	0.19	15.99	19.92	0.15
	RR	PCM_{RR} [54]	0.30	0.29	0.20	18.22	21.32	0.19
	NR	PQA-Net	0.70	0.69	0.52	13.45	15.27	0.50
Mushroom	FR	$PSNR_{MSE,p2po}$ [52]	0.66	0.63	0.48	12.92	16.27	0.53
		$PSNR_{MSE,p2pl}$ [17]	0.55	0.47	0.37	14.05	18.10	0.43
		$PSNR_{HF,p2po}$ [52]	0.48	0.26	0.20	14.84	18.97	0.34
		$PSNR_{HF,p2pl}$ [17]	0.45	0.22	0.16	15.59	19.35	0.23
		$PSNR_Y$ [53]	0.79	0.60	0.44	10.90	13.15	0.45
		AS_{Mean} [18]	0.39	0.26	0.18	15.83	19.99	0.29
		AS_{RMS} [18]	0.39	0.26	0.18	15.74	19.95	0.24
		AS_{MSE} [18]	0.39	0.26	0.18	15.74	19.95	0.24
		$SSIM_p$ [31]	0.85	0.73	0.57	8.76	11.48	0.59
		$MS\text{-}SSIM_p$ [33]	0.88	0.89	0.73	7.91	10.26	0.75
		$VIFP_p$ [32]	**0.90**	**0.90**	**0.76**	**6.70**	**9.42**	**0.78**
		$GraphSIM$ [26]	0.68	0.65	0.50	11.47	15.80	0.51
		$PCQM$ [22]	0.71	0.76	0.60	17.44	21.63	0.63
		$PointSSIM$ [37]	0.36	0.33	0.26	15.74	20.18	0.19
	RR	PCM_{RR} [54]	0.19	0.18	0.14	17.44	21.63	0.28
	NR	PQA-Net	0.77	0.71	0.56	11.29	13.81	0.56
Pineapple	FR	$PSNR_{MSE,p2po}$ [52]	0.52	0.41	0.30	13.89	16.85	0.32
		$PSNR_{MSE,p2pl}$ [17]	0.26	0.29	0.21	15.49	19.08	0.25
		$PSNR_{HF,p2po}$ [52]	0.29	0.11	0.09	15.31	18.89	0.22
		$PSNR_{HF,p2pl}$ [17]	0.34	0.17	0.12	14.68	18.53	0.17
		$PSNR_Y$ [53]	0.77	0.74	0.57	9.36	12.68	0.61
		AS_{Mean} [18]	0.24	0.25	0.20	14.84	19.18	0.24
		AS_{RMS} [18]	0.24	0.25	0.20	14.84	19.18	0.25
		AS_{MSE} [18]	0.23	0.25	0.20	14.86	19.19	0.25
		$SSIM_p$ [31]	0.82	0.82	0.66	8.97	11.26	0.63
		$MS\text{-}SSIM_p$ [33]	0.81	0.81	0.64	9.14	11.48	0.62
		$VIFP_p$ [32]	0.84	0.83	0.67	8.33	10.61	0.66
		$GraphSIM$ [26]	0.37	0.15	0.10	15.10	18.34	0.33
		$PCQM$ [22]	0.74	0.79	0.68	15.89	19.74	0.68
		$PointSSIM$ [37]	0.33	0.27	0.22	14.62	18.60	0.15
	RR	PCM_{RR} [54]	0.31	0.35	0.26	15.89	19.74	0.25
	NR	PQA-Net	**0.87**	**0.89**	**0.71**	**8.04**	**9.80**	**0.73**
Overall	FR	$PSNR_{MSE,p2po}$ [52]	0.50	0.48	0.33	15.08	18.43	0.36
		$PSNR_{MSE,p2pl}$ [17]	0.37	0.36	0.25	16.07	19.73	0.30
		$PSNR_{HF,p2po}$ [52]	0.34	0.16	0.11	16.42	20.00	0.11
		$PSNR_{HF,p2pl}$ [17]	0.27	0.20	0.14	16.85	20.48	0.16
		$PSNR_Y$ [53]	0.56	0.53	0.37	14.10	17.61	0.39
		AS_{Mean} [18]	0.24	0.24	0.16	16.83	20.65	0.21
		AS_{RMS} [18]	0.23	0.21	0.14	16.92	20.72	0.19
		AS_{MSE} [18]	0.23	0.21	0.14	16.91	20.72	0.19
		$SSIM_p$ [31]	0.59	0.58	0.42	13.98	17.23	0.41
		$MS\text{-}SSIM_p$ [33]	0.62	0.60	0.44	13.25	16.64	0.43
		$VIFP_p$ [32]	**0.82**	**0.82**	**0.63**	**9.85**	**12.23**	**0.62**
		$GraphSIM$ [26]	0.47	0.46	0.32	15.10	18.79	0.32
		$PCQM$ [22]	0.65	0.71	0.52	17.60	21.28	0.56
		$PointSSIM$ [37]	0.14	0.18	0.14	17.11	21.07	0.07
	RR	PCM_{RR} [54]	0.29	0.26	0.19	17.60	21.28	0.19
	NR	PQA-Net	0.70	0.69	0.51	12.36	15.18	0.50

there are too many features used in the metric, which limited its generalization ability. If the number of features is reduced sophisticatedly, and the weights are trained again on a larger dataset, the results should be better. The $PointSSIM$ is the

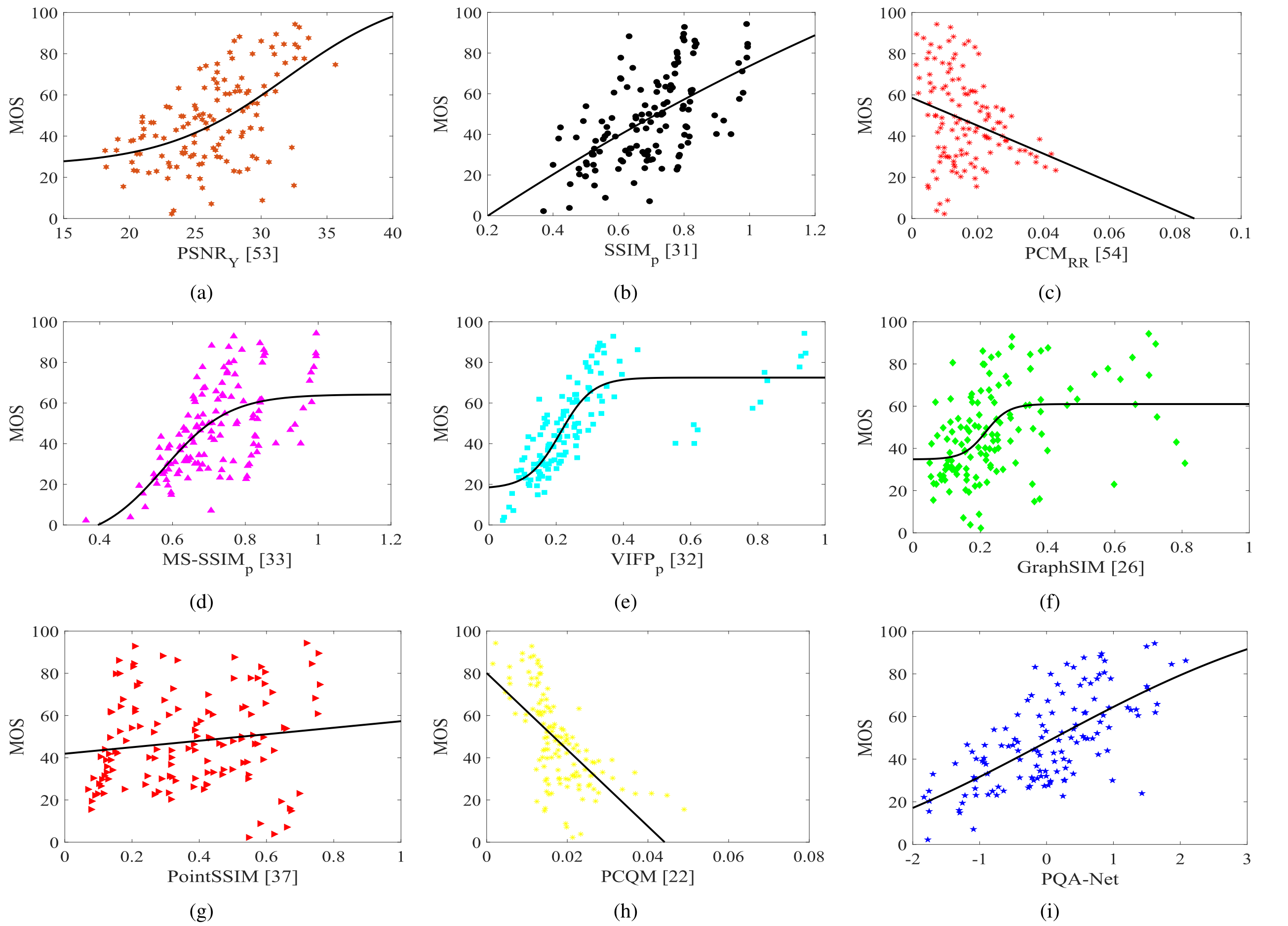


Fig. 8. Scatter plots of objective scores vs. MOSs. The best fitting logistic functions are also shown as solid curves. Note: the smaller the value of $PCQM$ and PCM_{RR}, the higher the MOS value of the distorted point cloud.

worst, whose overall PLCC and SRCC are 0.14 and 0.18, respectively.

All the projection-based methods, i.e., $SSIM_p$, $MS\text{-}SSIM_p$, and $VIFP_p$ achieve good performance, due to that both geometric and color information are taken into account, which is the same as the proposed PQA-Net. It is worth pointing out that the projected images employed in the proposed PQA-Net are the same with that used in the image-based metrics. We can see from Table III that $VIFP_p$ has the best performance, whose PLCC and SRCC are both as high as 0.82, and PQA-Net is the second best, whose PLCC and SRCC are as high as 0.70 and 0.69, respectively. Limited by the unfair usage of the original information, the performance of NR method is reasonably worse than the FR/RR methods. From the results, however, we can see that the performance of the proposed PQA-Net is only slightly lower than the best FR method $VIFP_p$, indicating the advantage of the proposed PQA-Net. From Table III we can also see that the prediction accuracy for the point cloud "*Banana*" is not good. The reason is that the point cloud "*Banana*" has a simple geometric structure and high-brightness single yellow colors, which are difficult for subjects to distinguish the quality changes. As a result, the quality labels of "*Banana*" for training are noisy, and the performance of PQA-Net is also reduced.

TABLE IV

THE CONFUSION MATRICES PRODUCED BY PQA-NET

Accuracy	Downsample	Noise	G-PCC	V-PCC
Downsample	**0.72**	0.38	0.00	0.00
Noise	0.00	**1.00**	0.00	0.00
G-PCC	0.00	0.00	**0.99**	0.01
V-PCC	0.00	0.00	0.51	**0.49**

To compare the accuracy of these quality metrics uniformly, a nonlinear four-parameter logistic function [60] is applied to map raw model predictions to the MOS scale. Scatter plots of objective scores and MOSs are shown in Fig. 8. The best fitting logistic functions are shown as solid curves in Fig. 8. All the compared methods have different score ranges from the MOSs, and their correspondence with MOSs are given by the fitting curve in each subplot of Fig. 8.

As a by-product, PQA-Net also outputs the distortion type of a point cloud. The confusion matrix [61] which is used to summarize the performance of a classification algorithm is shown in Table IV. The elements represent the probabilities that the distortion types are accurately predicted by the classifier. The higher the diagonal values of the confusion matrix, the better. Since the statistical behaviors of noise distortion have obvious distinction with the other three kinds of distortion, PQA-Net predicts noise distortion perfectly. On the

TABLE V

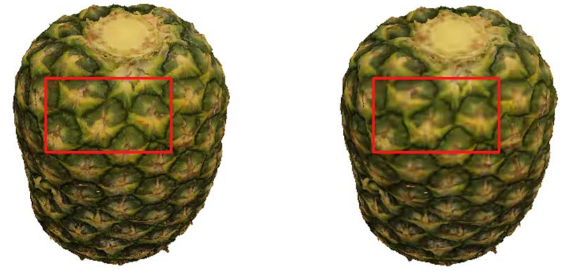
ACCURACY COMPARISON WITH EXISTING FR AND RR POINT CLOUD QUALITY METRICS FOR DIFFERENT DISTORTION TYPES

Distortion Type	Method Type	Method	$\|PLCC\|$	$\|SRCC\|$	$\|KRCC\|$
Downsample	FR	$PSNR_{MSE,p2po}$ [52]	0.96	0.86	0.67
		$PSNR_{MSE,p2pl}$ [17]	0.85	0.76	0.52
		$PSNR_{HF,p2po}$ [52]	0.97	0.85	0.64
		$PSNR_{HF,p2pl}$ [17]	0.97	0.84	0.61
		$PSNR_Y$ [53]	0.70	0.69	0.55
		AS_{Mean} [18]	0.96	0.77	0.52
		AS_{RMS} [18]	0.96	0.77	0.52
		AS_{MSE} [18]	0.96	0.77	0.52
		$SSIM_p$ [31]	0.91	0.87	0.73
		$MS\text{-}SSIM_p$ [33]	0.89	0.83	0.64
		$VIFP_p$ [32]	**0.98**	**0.91**	**0.76**
		$GraphSIM$ [26]	0.97	0.79	0.64
		$PCQM$ [22]	0.85	0.89	0.73
		$PointSSIM$ [37]	0.97	**0.91**	**0.76**
	RR	PCM_{RR} [54]	0.89	0.88	0.70
	NR	PQA-Net	0.97	0.80	0.64
Noise	FR	$PSNR_{MSE,p2po}$ [52]	0.67	0.63	0.44
		$PSNR_{MSE,p2pl}$ [17]	0.67	0.62	0.43
		$PSNR_{HF,p2po}$ [52]	0.67	0.63	0.45
		$PSNR_{HF,p2pl}$ [17]	0.67	0.63	0.45
		$PSNR_Y$ [53]	**0.90**	0.84	0.68
		AS_{Mean} [18]	0.71	0.68	0.49
		AS_{RMS} [18]	0.71	0.67	0.49
		AS_{MSE} [18]	0.70	0.67	0.49
		$SSIM_p$ [31]	0.84	0.66	0.48
		$MS\text{-}SSIM_p$ [33]	0.85	0.66	0.51
		$VIFP_p$ [32]	0.86	0.81	0.66
		$GraphSIM$ [26]	0.72	0.71	0.57
		$PCQM$ [22]	0.89	0.87	0.70
		$PointSSIM$ [37]	0.70	0.63	0.46
	RR	PCM_{RR} [54]	0.89	**0.88**	**0.73**
	NR	PQA-Net	0.75	0.64	0.44
G-PCC	FR	$PSNR_{MSE,p2po}$ [52]	0.41	0.39	0.29
		$PSNR_{MSE,p2pl}$ [17]	0.42	0.42	0.30
		$PSNR_{HF,p2po}$ [52]	0.40	0.42	0.30
		$PSNR_{HF,p2pl}$ [17]	0.34	0.34	0.23
		$PSNR_Y$ [53]	0.75	0.73	0.55
		AS_{Mean} [18]	0.13	0.03	0.08
		AS_{RMS} [18]	0.13	0.03	0.03
		AS_{MSE} [18]	0.13	0.03	0.03
		$SSIM_p$ [31]	0.63	0.63	0.48
		$MS\text{-}SSIM_p$ [33]	0.67	0.66	0.51
		$VIFP_p$ [32]	**0.84**	0.83	0.65
		$GraphSIM$ [26]	0.70	0.61	0.44
		$PCQM$ [22]	0.72	**0.85**	**0.69**
		$PointSSIM$ [37]	0.78	0.77	0.60
	RR	PCM_{RR} [54]	0.20	0.19	0.13
	NR	PQA-Net	0.68	0.67	0.51
V-PCC	FR	$PSNR_{MSE,p2po}$ [52]	0.48	0.42	0.30
		$PSNR_{MSE,p2pl}$ [17]	0.48	0.46	0.32
		$PSNR_{HF,p2po}$ [52]	0.35	0.27	0.19
		$PSNR_{HF,p2pl}$ [17]	0.52	0.43	0.32
		$PSNR_Y$ [53]	0.48	0.32	0.25
		AS_{Mean} [18]	0.66	0.53	0.35
		AS_{RMS} [18]	0.60	0.49	0.32
		AS_{MSE} [18]	0.60	0.49	0.32
		$SSIM_p$ [31]	0.44	0.35	0.29
		$MS\text{-}SSIM_p$ [33]	0.50	0.38	0.34
		$VIFP_p$ [32]	**0.91**	**0.90**	**0.76**
		$GraphSIM$ [26]	0.61	0.32	0.27
		$PCQM$ [22]	0.59	0.59	0.45
		$PointSSIM$ [37]	0.51	0.48	0.34
	RR	PCM_{RR} [54]	0.50	0.55	0.42
	NR	PQA-Net	0.60	0.45	0.30

other hand, the compression distortion generated by G-PCC and V-PCC is more easily to be confused by the classifier. Since the number of samples with G-PCC distortion is about 1.33 times more than the V-PCC, the classification accuracy of G-PCC is better. Accordingly, we can speculate that using a larger dataset is likely to result in better performance. To measure the performance of classifier, the accuracy of distortion type classifier was calculated. The accuracy is defined as the proportion of the total number of predictions that are correct:

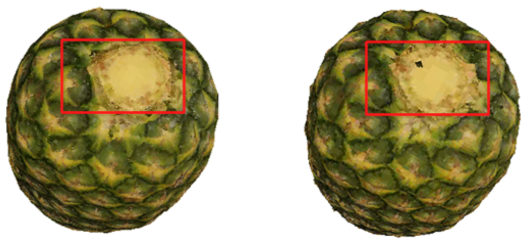
$$DT_a = \frac{DT_r}{DT_t}, \tag{11}$$

where DT_r denotes the number of correct predictions made by a distortion type classifier, and DT_t denotes the total



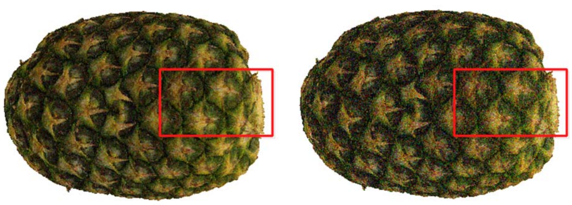
(a) Two point clouds with the same quality predicted by $PointSSIM$ [37]

(b) Two point clouds with the same quality predicted by $GraphSIM$ [26]



(c) Two point clouds with the same quality predicted by $PCQM$ [22]

(d) Two point clouds with the same quality predicted by $PSNR_Y$ [53]



(e) Two point clouds with the same quality predicted by MS-$SSIM_p$ [33]

(f) Two point clouds with the same quality predicted by $SSIM_p$ [31]

(g) Two point clouds with the same quality predicted by $VIFP_p$ [32]

(h) Two point clouds with the same quality predicted by PQA-Net

Fig. 9. Subjective comparison between two point clouds which have the same quality predicted by evaluated quality metrics.

number of all the tested point clouds. From Table IV, we can compute the mean DT_a predicted by PQA-Net is as high as $(0.72 + 1.00 + 0.99 + 0.49) \div 4 = 0.8$.

4) Experimental Results on Distortion Types: We also compared the proposed PQA-Net with classic and state-of-the-art FR and RR point cloud quality metrics for different distortion types.

These results are shown in Table V. First of all, for the $downsample$ distortion type, the performance of all methods is relatively good. Since this kind of distortion is caused by directly discarding some points of the original point cloud, which damage the geometry and color information rudely. This is also the reason why the methods (i.e., $PSNR_{MSE,p2p0}$, $PSNR_{MSE,p2pl}$, $PSNR_{HF,p2p0}$, $PSNR_{HF,p2pl}$, AS_{Mean}, AS_{RMS}, and AS_{MSE}) that only consider the geometry structure also perform good. For the $noise$ distortion type, white gaussian noise was added independently to both geometry and color elements, respectively. $PSNR_Y$ performs the best since it focuses on the color noise at the corresponding point perfectly. For the G-PCC and V-PCC distortion types which

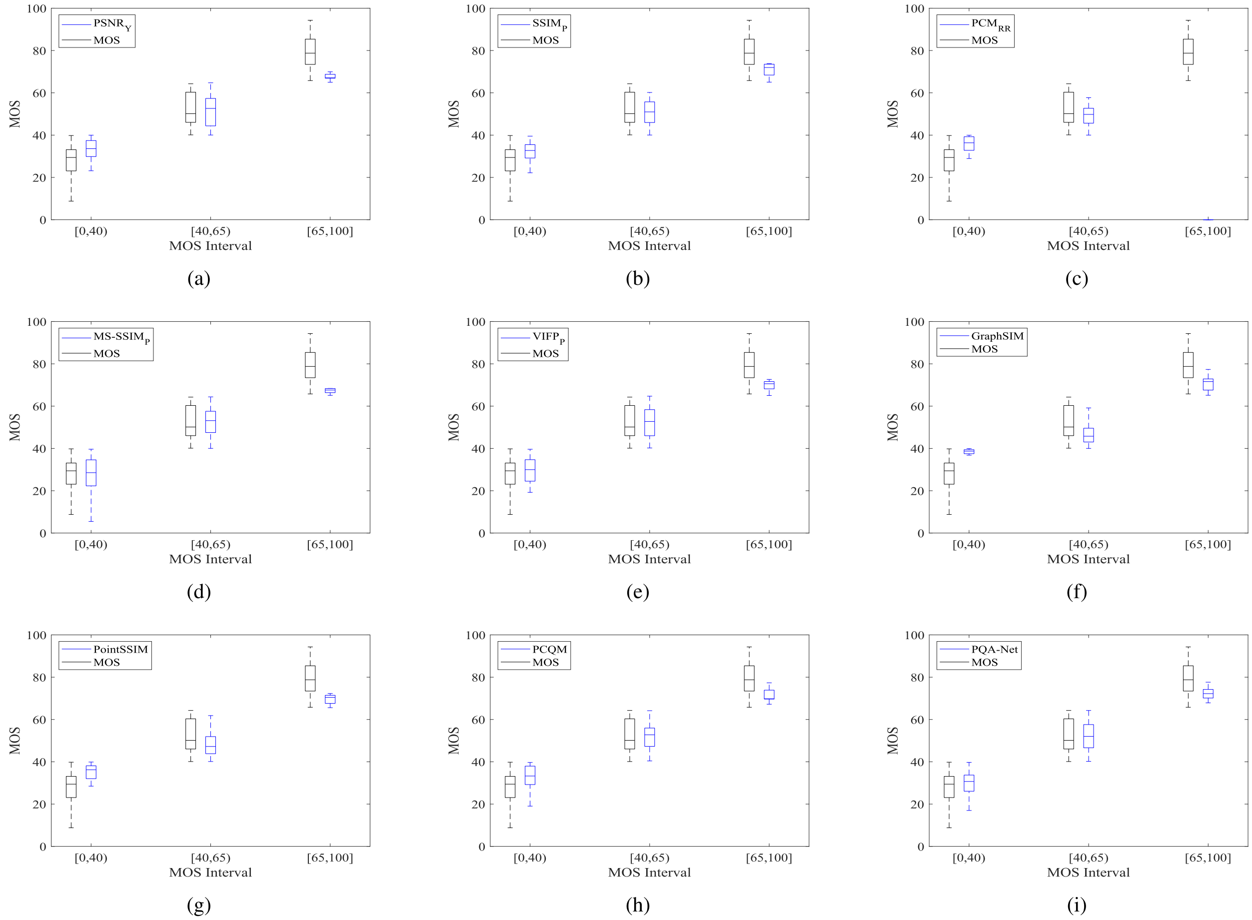


Fig. 10. Box plot visualization of MOS intervals.

TABLE VI

INFLUENCE OF THE DIFFERENT LOSS FUNCTIONS $\widehat{l_2}$ IN (12)

$\widehat{l_2}$ loss function	PLCC	SRCC	KRCC	MAE	RMSE
L1	0.54	0.62	0.44	14.49	17.97
L2	0.42	0.44	0.30	15.87	19.27
Smooth L1	0.45	0.45	0.32	15.38	19.03

introduces both geometry and color coding distortions, the projection-based methods are better than the point-based and the angular-based methods (as mentioned in Sec. IV-C3) due to comprehensive consideration of distortion of geometry and color. In summary, $VIFP_p$ performs the best for all distortion types. PCM_{RR}, $PCQM$, and $PointSSIM$ also perform well in specific distortion types but they are not robust enough. As a comparison, the performance of the proposed PQA-Net is robust enough, and its accuracy is very close to the FR metric $VIFP_p$.

5) Visual Result: As we know, human vision is the ultimate standard to evaluate different quality metrics [62]. Accordingly, we further compared the consistency between the visual results of PQA-Net and the state-of-the-art methods. Since the distribution ranges of the point cloud quality scores predicted by different evaluation methods are very different, we compared the visual difference of two degraded point clouds which have the same predicted quality score obtained from the evaluated quality metrics. It is worth mentioning that these two degraded point clouds are selected from the same distortion type for fair comparison. Take the subjective visual difference of the point cloud $pineapple$ as an example, as shown in Fig. 9, we can see that the visual perception of the two degraded point clouds is very different in Fig. 9. (a), but the $PointSSIM$ predicts that the quality of the two degraded point clouds are the same, indicating the quality metric $PointSSIM$ is inaccurate. Similar results can also be found in Fig. 9. (b), (c), (d), and (f) for $GraphSIM$, $PCQM$, $PSNR_Y$, $MS\text{-}SSIM_p$, $SSIM_p$. The visual perceptions of the two degraded point clouds in Fig. 9. (g) and Fig. 9. (h) are very close, and the quality scores predicted by the $VIFP_p$ and the proposed PQA-Net are also very close, indicating that the prediction quality score evaluated by the $VIFP_p$ (an FR method) and the proposed PQA-Net (a NR method) are consistency with the subjective vision. To better compare the trend similarity between the predicted MOSs of each quality metric and the ground truth, we divided MOS interval into three sub-intervals ([0, 40), [40, 65) and [65, 100], respectively), and then compared the box plot statistical diagrams, as shown in Fig 10. Note that we applied a nonlinear four-parameter logistic function [60] to map raw model predictions to the MOS scale in Fig 10. It can be seen that the trend of the median line of the predicted MOSs of our proposed PQA-Net is very similar to that of the trend of the ground truth MOSs. The median line of the predicted MOSs of PCM_{RR} is also close to that of the ground truth MOSs in sub-intervals [0, 40)

TABLE VII
INTRODUCTION OF OTHER DATASETS

Datasets	Degradation Number	Distortion Type	Content Number	MOS range
IRPC [65]	54	PCL, G-PCC, V-PCC	6	[1, 5]
M-PCCD [66]	232	Octree-Lifting, Octree-RAHT, TriSoup-Lifting, TriSoup-RAHT, V-PCC	8	[1, 5]
SJTU-PCQA [36]	144	Octree-based Compression, Color Noise, Downscaling, Geometry Gaussian Noise	6	[1, 10]

and [40, 65), but unfortunately, there are no any raw model predictions mapped into [65, 100] interval.

6) Ablation Studies for Loss Functions: We investigated more loss functions for PQA-Net to explore the performance.

The widely used loss functions, e.g. L1 loss which measures the mean absolute error between each element in the input objective and target objective [14], L2 loss which measures the mean squared error between each element in the input objective and target objective [63], and smooth L1 loss which uses a squared term if the absolute element-wise error falls below a threshold and an L1 term otherwise [64], are chosen. In this ablation study, we adjusted the hybrid loss function of PQA-Net as

$$l(\mathbf{X}^{(k)}; \mathbf{W}, \mathbf{w}_1, \mathbf{w}_2) = l_1 + \lambda_{ablation}\widehat{l_2}, \tag{12}$$

where $\lambda_{ablation}$ is set to be 0.025 to balance the difference between the loss of distortion classifier and quality predictor task, l_1 is the cross entropy to measure the classification error of sub-task I, $\widehat{l_2}$ is chosen from L1, L2, or Smooth L1 to measure the error between the actual and predicted scores for sub-task II, and $\mathbf{W}, \mathbf{w}_1, \mathbf{w}_2$ are the parameters of MVFEF, DTI, and QVP modules, respectively. The result of different loss functions are given in Table VI. We can see that the highest PLCC and SRCC of L1/L2/Smooth L1 loss functions are 0.54 and 0.62, respectively, and the lowest MAE is 14.49. However, the average PLCC and SRCC of the PLCC-based loss function (5) are 0.70 and 0.69, respectively, and the MAE is 12.36 (see Table III), indicating its advantage.

7) Ablation Studies on Different Datasets: To fully test the performance of PQA-Net, we also conducted further experiments on the other subjective datasets. The subjective datasets are briefly introduced in Table VII. For SJTU-PCQA dataset, as there are some distorted point clouds with mixed distortion types which are useless for our testing, only the distorted point clouds with individual distortion types are tested. In the experiments, the weighting factor λ in the proposed hybrid loss function (5) was set to be 15 for IRPC [65], 15 for SJTU-PCQA [36], and 25 for M-PCCD [66] to adapt to the different distortion types of these dataset. We randomly choose 80% reference point clouds for training and the rest 20% reference point clouds are left out for testing on each dataset. There are no overlapping samples between the training and the testing sets.

Detailed experiment results are given in Table VIII, from which we can see that the performance on the IRPC dataset is the worst. This is because the amount of training data of the IRPC dataset is very small, which greatly limits the performance of the proposed PQA-Net. The performance of PQA-Net on M-PCCD is better, but is not yet perfect. This is because the differences between the distortion types are small, especially for the distortion types of Octree-Lifting, Octree-RAHT, TriSoup-Lifting, and TriSoup-RAHT, which were all generated from the G-PCC encoder. To distinguish these similar distortion types is difficult for humans, let alone a neural network based on limited training dataset. The performance of PQA-Net is the best on SJTU-PCQA dataset because the difference between the distortion types on SJTU-PCQA dataset is obvious.

TABLE VIII
THE PERFORMANCE OF PQA-NET ON DIFFERENT DATASETS

Datasets	PLCC	SRCC
SJTU-PCQA [36]	0.85	0.82
IRPC [65]	0.58	0.40
M-PCCD [66]	0.60	0.65

V. CONCLUSION

We proposed a deep learning-based NR point cloud quality assessment method, namely PQA-Net, which contains feature extraction, distortion type identification, and the quality vector prediction modules. The most important advantage of proposed PQA-Net is that it is the first NR quality assessment method for point clouds, which is crucial for point cloud communication system. In addition, DNN is first introduced for point cloud quality assessment, and the limitation of small sample data was also overcome. Experimental results demonstrated it can achieve comparable or even better performance than most of the existing FR and RR point cloud quality metrics.

One potential future direction is to reasonably and effectively expand the subjective dataset to further improve the performance. Another potential direction is to optimize the network architecture by introducing advanced neural modules.

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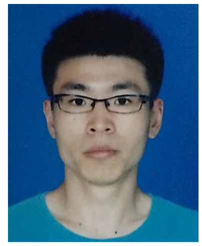

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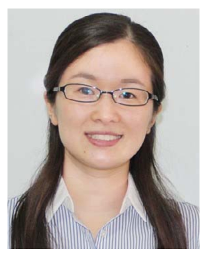

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